

Data Science and Ethical Issues

Lecture 8



Learning Outcomes

- Ethical and privacy implications of collecting and managing big data
- Discussions on data ownership
- Data privacy, Security, Anonymity, Data validity
- Societal consequences of data science.
- Real datasets and applications from a variety of disciplines will be used.



Ethics and Privacy Issues

- Employers search social networking sites for background information on applicants
- Monitoring systems are adopted by employers to check employee performance
- Information technology has aided in the easy availability of personal information
 - Information in databases can be used for direct marketing and for credit checks on potential borrowers or renters



Concepts of the Web and Network Privacy

Acceptable use policy

- Set of rules specifying the legal and ethical use of a system and the consequences of noncompliance

Accountability

- Issues involving both the user's and the organization's responsibilities and liabilities

Nonrepudiation

- Method for binding all the parties to a contract



Guidelines to Minimize the Invasion of Privacy

- Conduct business only with websites having privacy policies
- Limit access to personal information to those with authorization
- Ensure data's reliability and take precautions to prevent misuse of the data
- Data collection must have a stated purpose
- Identify ways to prevent personal information gathered being disclosed without consent



Guidelines to Minimize the Invasion of Privacy

- Organizations should use verification procedures to ensure data accuracy
- Organizations must correct or delete incorrect data
 - Prevent unauthorized access to data and misuse of data
- Users should review their records and correct any inaccuracies
- Transparency in record-keeping systems storing personal data



Privacy Issues Presented by E-mail

- **Spam:** Unsolicited e-mail sent for advertising purposes
 - Sent in bulk using automated mailing software
- **Ease of access**
 - Individuals should assume that others have access to their messages
 - Any e-mails sent on organization-owned computers are the property of the organization



Data Collection on the Web

- Due to the increase in online shopping, personal details of individuals are prone to misuse
 - Issue of personal information being sold to telemarketing firms resulting in spam
- Information provided on the Web can be combined with other information and technologies to produce new information



Technologies Used for Data Collection

- **Cookies:** Small text files with unique ID tags that are embedded in a Web browser and saved on the user's hard drive
 - Help websites customize pages for users
 - Considered an invasion of privacy when users' information is used without prior consent
 - Installing a cookie manager helps users disable cookies
- **Log files:** Generated by Web server software, record a user's actions on a website



Ethical Issues of Information Technologies

- Information technology offers opportunities for unethical behavior due to ease of collecting and disseminating information
 - Increase of cybercrime, cyberfraud, identity theft, and intellectual property theft
- Organizations can reduce unethical behavior of employees by developing and enforcing codes of ethics



Ownership and Censorship of Information on the Web

- **Public information**

- Posted by an organization or public agency
- Censored for public policy reasons
- Censored if the content is deemed offensive to a political, religious, or cultural group

- **Private information**

- Posted by a person
- Uncensored because of constitutional freedom of expression



Censorship

- **Restricting access to the Web**

- Countries like Burma, China, and Singapore, restrict or forbid their citizens' access to the Web
 - Internet neutrality: Internet service providers (ISPs) and government agencies should treat all data on the Internet equally
- Parents use programs like CyberPatrol, CyberSitter, Net Nanny, and SafeSurf to prevent children's access to certain websites



Intellectual Property

- Legal umbrella covering protections that involve copyrights, trademarks, trade secrets, and patents developed by people or businesses



Categories in Intellectual Property

- **Industrial property**
 - Inventions, trademarks, logos, and industrial designs
- **Copyrighted material**
 - Literary and artistic works
 - Covers online materials like Web pages, HTML code, and computer graphics
 - Fair Use Doctrine: Exception to the Copyright law which allows the use of copyrighted material for certain purposes



Intellectual Property Protections

- **Trademark:** Protects product names and identifying marks
- **Patent:** Protects new processes
 - Advantages to organizations
 - Generates revenue by licensing the patent
 - Attracts funding for research and development
 - Keeps competitors from entering certain market segments



Laws Covering Software Piracy

- 1980 revisions to the Copyright Act of 1976 include computer programs
- Laws covering legal issues related to information technologies in the United States
 - Telecommunications Act of 1996
 - Communications Decency Act (CDA)
 - Laws against spamming



Cybersquatting

- Registering, selling, or using a domain name to profit from someone else's trademark
- Typosquatting: Relies on typographical errors made by Web users when typing a website address into a Web browser
 - Variation of cybersquatting
 - Called URL hijacking



Digital Divide

- Information technology and the Internet that is not accessible for all
 - Increasing funding for computers at schools, public places, helps offset the divide



Impact of Data Science in the Workplace

- Creation of technical jobs, requiring extensive training
- Increased consumers' purchasing power
 - Results in a stronger economy by reducing production costs
- Direct effect on the nature of jobs
 - Telecommuting enables work from home option
 - Organizations can use the best and most cost-effective human resources in a large geographical region



Impact of Data Science in the Workplace

- Job deskilling: Occurs when skilled labor is eliminated due to high technology
- **Virtual organizations:** Companies, suppliers, customers, and manufacturers connected via information technologies
 - To share skills and cost
 - To have access to each other's markets



Data Science and Health Issues

- Increasing popularity of touchscreens result in stress related injuries of the users' hands, arms, back, and eyes
- Related health problems associated with computer equipment
 - Vision issues
 - Musculoskeletal issues
 - Skin issues
 - Reproductive system issues
 - Stress issues



Green Computing

- Promotes a sustainable environment and consumes the least amount of energy
 - Involves the design, manufacture, use, and disposal of computers, servers, and computing devices with minimal impact on the environment
- Requires cooperation of both private and public sectors



Ways to Achieve Green Computing

- Design products that last longer and are modular in design
 - Parts can be upgraded without replacing the entire system
- Design search engines and computing routines that are faster and consume less energy
- Replace underutilized smaller servers with one large server using a virtualization technique



Ways to Achieve Green Computing

- Use computing devices that consume less energy and are biodegradable
- Allow certain employees to work from their homes to reduce carbon foot prints
- Conduct meetings over computer networks to reduce business travel
 - Encourage carpool and nonmotorized transportation for employees
- Turn off idle PCs and recycle computer-related materials



Key Terms

- **Acceptable use policy**
- **Accountability**
- **Cookies**
- **Cybersquatting**
- **Digital divide**
- **Green computing**
- **Intellectual property**
- **Log files**
- **Nonrepudiation**
- **Spam**
- **Virtual organizations**



Summary

- Employers search social networking sites, such as Facebook or MySpace, to find background information on applicants
- e-mail is distributed through the Web or a company network; people should assume that others have access to their messages
- Public information, posted by an organization or public agency, can be censored for public policy reasons
- Children, in particular, are victims of the digital divide
- Information technology has created many new jobs
- Green computing is one of the methods for combating global warming